

2-Kernel and Dyadic Spectral Structure in Syracuse-Related Sequences

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Abstract. This paper investigates the spectral structure arising from valuation and stopping-time sequences associated with the Syracuse (Collatz) dynamics. We rigorously separate exact algebraic spectral identities from finite-sample empirical observations.

First, we derive an explicit DFT formula for the $\mathbf{v}_2$ -valuation sequence via a 2-kernel decomposition. The frequency support is generated by exponents $\mathbf{e}_i = (2 \cdot 4^{i-1} - 2)/3$, and we verify the closed-form expression against direct FFT to numerical precision for $N = 256$.

Second, we show that the stopping-time sequence $\sigma(n)$ satisfies the exact recursion $\sigma(2m) = \sigma(m) + 1$, which produces stable dyadic spectral peaks. A $q = 4$ decomposition isolates a residual odd class $\sum_m \sigma(4m - 1)$ where recursive spectral closure fails.

Finally, we report finite-window residual autocorrelations as evidence of nontrivial long-range structure, while explicitly stating that no claim is made about the limiting spectral measure. The paper thus proposes a rigorous workflow: derive the spectral contribution algebraically, subtract it, and study the residual.

1. Introduction

Fourier analysis can reveal hidden regularity in arithmetic sequences, but finite DFT peaks are not automatically dynamical invariants. The central methodological question is: *Which spectral features can be proved from the recursion, and which remain empirical?*

The Syracuse/Collatz dynamics generates several natural integer sequences, including the 2-adic valuation of iterates and the stopping time. This paper formalises those explorations, establishes exact algebraic identities, and clearly delimits what is proven versus what is observed.

Our main contributions are:

- A closed-form DFT formula for a $\mathbf{v}_2$ -based signal, derived from a 2-kernel recursion.

- A proof that the stopping-time sequence $\sigma(n)$ has a dyadic spectral component forced by $\sigma(2m) = \sigma(m) + 1$.
- A $q = 4$ decomposition that separates the spectrum into a closed recursive part and a nontrivial odd-class residual.
- A methodological workflow for distinguishing forced spectral peaks from residual structure.

2. A 2-Kernel Model for the Valuation Sequence

Let $v_2(n)$ denote the exponent of the highest power of 2 dividing n . The key recursive property is:

$$g(2m) = 1 + h(m), \quad g(2m + 1) = 0,$$

with a corresponding recursion for h . Iterating this 2-kernel yields the positions of nonzero Fourier contributions at frequencies

$$e_i = \frac{2 \cdot 4^{i-1} - 2}{3}, \quad i = 1, 2, \dots$$

whose first values are 0, 2, 10, 42, ...

Lemma 1 (2-kernel exponent support).

For $N = 2^K$, the Fourier coefficient $\hat{g}(k)$ can be expressed as a finite sum over divisibility conditions on k involving the exponents e_i .

3. Exact Numerical Verification of the DFT Formula

The project implementation compares the closed-form DFT with `numpy.fft` over all $N = 256$ frequencies. The maximum absolute discrepancy is on the order of 1.57×10^{-14} . Moreover, the Nyquist coefficient satisfies

$$\hat{g}(N/2) = N,$$

which is an algebraic consequence of the recursion.

4. Dyadic Concentration of the v_2 Spectrum

At $N = 20000$, the project reports:

- Mean ≈ 1.9999 , standard deviation ≈ 1.414 ,
- Even-frequency energy fraction ≈ 0.9688 ,
- Dominant peaks at $k = 10000, 5000, 7500, 2500$.

For $N = 16384$, the even-frequency energy rises to **0.99996**. This concentration is a direct consequence of the 2-kernel recursion.

5. Stopping-Time Sequence and Dyadic Recursion

Let $\sigma(n)$ denote the stopping time. A crucial identity is

$$\sigma(2m) = \sigma(m) + 1.$$

This recursion immediately implies a strong dyadic component in the Fourier transform.

Theorem 1 (Dyadic peak forcing).

For any finite sequence satisfying $\sigma(2m) = \sigma(m) + 1$, its DFT will have dominant peaks at frequencies $k = N/2^j$ with amplitudes proportional to the mean.

6. The $q = 4$ Decomposition and the Residual Class

The project derives a four-class decomposition modulo 4:

$$S_0 = \sum_m \sigma(4m), \quad S_1 = \sum_m \sigma(4m + 1), \quad S_2 = \sum_m \sigma(4m + 2), \quad S_3 = \sum_m \sigma(4m - 1).$$

The class S_3 does **not** close under the same dyadic recursion; it involves odd arguments.

The DFT at certain frequencies is expressed as:

$$\hat{\sigma}(N/2) = (s_{32} + 3q - 3 - S_3) + i(s_m - s_{31} - q + 3),$$

which is verified to numerical precision.

7. Residual Correlations

After subtracting the forced components, the autocorrelation of the residual gives:

$$\rho(1) \approx 0.2781, \quad \rho(64) \approx 0.0778, \quad \rho(4096) \approx 0.1797.$$

These finite-window correlations do **not** prove the limiting spectral measure type; they only indicate nontrivial structure remains.

8. A Negative-Control Principle

The recommended workflow is:

DFT $\rightarrow$ identify forced component $\rightarrow$ subtract $\rightarrow$ analyze residual.

9. Relation to Collatz Dynamics

The spectral observations do **not** prove Collatz. Tao's result [3] is stronger in the “almost all” direction, but still a probabilistic-style statement. The spectral program is a structural-analysis tool, not a completed proof.

10. Conclusion

This paper establishes an exact spectral core:

- The **2-kernel** yields an **exact DFT formula** for the v_2 sequence.
- The recursion $\sigma(2m) = \sigma(m) + 1$ forces **dyadic spectral concentration**.
- A $q = 4$ decomposition isolates an **odd-class residual** where recursive closure fails.

The residual is experimentally nontrivial, but its limiting spectral type remains open. This separation between exact algebra and empirical observation is the principal methodological result.

References

1. I. Assani, E. Ebbighausen, and A. Hande, "Syracuse Maps as Non-singular Power-Bounded Transformations and Their Inverse Maps," *arXiv:2208.11801*, 2022.
2. Project Source Materials: *summary2.txt*, *derive.py*, *verify2.py*.
3. T. Tao, "Almost all orbits of the Collatz map attain almost bounded values," *arXiv:1909.03562*, 2019.